

Manan Rastogi

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Applied AI Engineer

Computer Vision | Full-Stack Systems | LLM Integration

Education

- 2024–Present **B.Tech in Electronics & Communication Engineering**, Vellore Institute of Technology (VIT), Vellore, India, CGPA: 7.53
Biomedical Specialization
- 2025–Present **BS in Data Science & Applications**, IIT Madras, Online, India

Internship

- May 2026–Jun 2026 **Summer Intern - Retail Automation**, Indian Oil Corporation, Bareilly, India
- Designed and developed an automated SLA compliance monitoring and penalty calculation tool for retail automation operations.
 - Automated tracking of service-level agreement performance against agreed norms, replacing manual monitoring workflows.
 - Implemented penalty computation logic with configurable rules and documented the tool for the operations team.

Technical Skills

- Languages** Python, TypeScript, JavaScript
- AI/ML & CV** OpenCV, SciPy, NumPy, TrackPy, OCR, LLM APIs (NVIDIA Llama 3.1)
- Frontend** React, Next.js, Three.js, Tailwind CSS, Framer Motion
- Back-end/DBs** Flask, Node.js, REST APIs, JWT, MongoDB, SQLite, SQL
- Infra/Tools** Docker, CI/CD, Vercel, MongoDB Atlas, Git, GitHub, VS Code

Independent Development

- 2024–Present **WoundTrack AI: Applied Computer Vision Pipeline**
Technologies: Python, OpenCV, Flask, SciPy, SQLite [Live Demo](#) | [GitHub](#)
- Engineered a CV pipeline tracking 441 cells across 800+ frames in <10 min on consumer hardware, achieving 28% higher precision than manual ImageJ workflows.
 - Designed LoG blob detection + LAP gap-closing tracker with Kalman-filter prediction, bridging detection gaps to reduce trajectory fragmentation during cell occlusions.
 - Built automated evaluation pipeline computing MSD exponents, velocity distributions, and fractal metrics across 5 experimental conditions.
 - Shipped Flask + SQLite dashboard auto-generating per-experiment reports with kinetic curves and one-click trajectory exports, replacing manual spreadsheet analysis.
- 2025–Present **VITAlize AI: Full-Stack Academic Platform**
Technologies: Next.js 15, TypeScript, MongoDB, REST APIs, JWT, NVIDIA LLAMA 3.1, Tesseract OCR [Live Demo](#)
- Developed a unified academic platform indexing 8,800+ past exam papers with full-text search and subject-level filtering across FFCS planner, paper vault, and exam generator modules.
 - Integrated NVIDIA LLAMA 3.1 with Tesseract OCR fallback pipeline to auto-generate mark-weighted AI solutions from scanned exam PDFs with KaTeX-rendered math.
 - Engineered constraint-based FFCS timetable solver using pruned backtracking algorithm, resolving clash/faculty/gap constraints across 2,400+ course combinations with sub-second solve time.
 - Deployed on Vercel with MongoDB Atlas; serverless architecture handling concurrent academic workloads. Built Chrome extension auto-syncing academic data from VIT's portal.
- 2024–Present **Interactive 3D Engineering Portfolio**
Technologies: TypeScript, React, Three.js (R3F), Framer Motion, Tailwind CSS [Live Site](#) | [GitHub](#)
- Architected a React + TypeScript portfolio with lazy-loaded bundle splitting, sub-1s initial load, and responsive accessibility-first layout.
 - Implemented WCAG accessibility: keyboard navigation, ARIA labels, skip-to-content, reduced-motion support, and focus management (Lighthouse: 100 SEO, 97 accessibility, 100 best practices).
 - Reduced layout shift and animation overhead by implementing Framer Motion with CSS-based reduced-motion fallbacks while maintaining 60 FPS on mobile devices.